

Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 7

Deadline: by the end of next Wednesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems. **Justify your answers.**

Problem 1. Determine the order and sign of each of the following permutations:

- (i) $(1\ 2\ 3\ 4\ 5)(8\ 7\ 6)(10\ 11)$;
- (ii) $(1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10)$;
- (iii) $(1\ 2)(3\ 4)(5\ 6\ 7\ 8)(9\ 10)$;
- (iv) $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)(1\ 8\ 7\ 6\ 5\ 4\ 3\ 2)$.

Problem 2. Show that every element of S_n ($n \geq 2$) is a product of transpositions of the form $(k\ k+1)$.

Problem 3. Decide which of the following sets are groups under the given operations:

- (v) The set of all real numbers of the form $a + b\sqrt{2}$, where a and b are integers, under addition.
- (vi) The set of all 3×3 matrices of the form

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix},$$

where a, b, c are real numbers, under matrix multiplication.

- (vii) The set of integers under subtraction.
- (viii) The set of real numbers under the operation $*$ defined by

$$a * b = a + b + 2.$$

Problem 4. Let $(G, *)$ be a group, and let H be a non-empty subset of G . Prove that H , with the operation $*$ restricted to H , is a group if and only if

$$h * k^{-1} \in H$$

for all $h, k \in H$.

Problem 5. Complete the partially given group table below. The identity element need not be the first column heading.

	a	b	c	d	f	g
a	c			f		
b		f			c	
c	a					
d				c		
f						
g		d				c